

Hardik Santosh Gaonkar

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AI & ML undergraduate building LLM systems end to end — RAG, AI agents, ML and computer vision — and proving they work with held-out benchmarks, confidence intervals and CI quality gates. Seeking AI/ML engineering internships.

EDUCATION

BMS Institute of Technology and Management, Bengaluru	2024 – 2028 (expected)
B.E. in Artificial Intelligence and Machine Learning	CGPA: 8.09 / 10
St. Joseph College of Commerce and Science — Class XII (PUC)	93.6%
Balmandir High School — Class X	97.6%

PROJECTS

CareFlow AI | Clinical AI platform with access-controlled RAG, ML and X-ray triage github.com/hrdk6/CareFlow-AI

Python · FastAPI · PostgreSQL + pgvector · scikit-learn · XGBoost · SHAP · ONNX Runtime · Next.js · TypeScript · Docker

- Built a full-stack clinical AI platform where the LLM only sees data the signed-in user may access: RBAC and row-level SQL filters run before retrieval, ML or generation, and a deterministic router plans SQL, RAG and ML tool calls.
- Implemented hybrid RAG (bge-small embeddings + BM25, RRF fusion, cross-encoder reranking, citation validation), reaching **Recall@1 0.955** and **MRR 0.977** on a 48-question benchmark; reranking lifted context precision from 0.62 to 0.84.
- Trained explainable risk models on 99K patient encounters: a calibrated Random Forest for 30-day readmission (**PR-AUC 2× base rate**, SHAP) and XGBoost length of stay, with patient-grouped splits and leakage ablations.
- Applied transfer learning to chest X-ray triage: logistic heads on frozen ResNet-50 (ONNX) features, trained on 26K NIH ChestX-ray14 films; **8 findings reached ROC-AUC 0.71–0.80** on 7.5K patient-disjoint test films, with class activation maps.
- Created a 3-pass privacy gateway (identifier vault, regex, ONNX BERT NER) that pseudonymizes patient data before prompts reach cloud LLMs and restores it server-side, with Groq → Gemini → extractive failover; **269 automated tests**.

PatchPilot | Autonomous bug-fixing coding agent with sandboxed execution github.com/hrdk6/PatchPilot

Python · LangGraph · FastAPI · SQLAlchemy · Qdrant · Docker · React · TypeScript · GitHub Actions

- Developed an agent that turns a repository and a bug report into a tested patch: a 9-node **LangGraph** state machine (index → retrieve → plan → patch → sandbox test → repair) whose budget-capped repair loop feeds test failures back to the LLM.
- Designed explainable code retrieval: AST-based, symbol-level chunking and hybrid ranking over **7 weighted signals** (semantic, lexical, import/call graph, failure trace, ...), logging why each chunk entered the prompt.
- Sandboxed LLM-written code in hardened Docker (no network, non-root, all capabilities dropped, CPU/memory/PID/time limits) behind a patch policy that blocks CI configs, lockfiles and path traversal, and redacts secrets from output.
- Built a model-agnostic benchmark harness (OpenAI, Anthropic, Ollama) reporting pass rate with **Wilson 95% CIs**, p50/p95 latency, tokens and cost; **421 tests** and a 5-job GitHub Actions pipeline that boots the full Docker stack.

GroundTruth | Self-verifying RAG with a rigorous evaluation harness github.com/hrdk6/GroundTruth

Python · FastAPI · PostgreSQL + pgvector · Sentence-Transformers · Claude / NVIDIA NIM · Next.js

- Engineered version-aware RAG over multi-release Kubernetes docs: pgvector HNSW + **BM25 implemented in SQL** (matches a Python reference to 1e-9), RRF fusion, query rewriting and multi-hop decomposition.
- Verified every generated sentence against its cited source, regenerating once or abstaining: **faithfulness 0.94–0.97**, citation precision 0.90–1.00 on dev/test, and 100% of unanswerable questions declined.
- Evaluated every change with bootstrap 95% CIs, paired significance tests and a CI regression gate; a self-audit caught **13 measurement bugs**, including one that produced a spurious +0.50 recall@5 gain.
- Dropped cross-encoder reranking after measuring ~100× latency for no significant recall gain; added per-stage tracing, six-way failure attribution and a Next.js trace viewer; **256 tests**.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
GenAI & LLMs	RAG, hybrid search (BM25 + dense), reranking, embeddings, LLM agents, LangGraph, tool calling, prompt engineering, LLM evaluation, guardrails (PII masking, prompt injection)
LLM Platforms	Anthropic Claude, OpenAI, Google Gemini, Groq, NVIDIA NIM, Ollama
ML, NLP & CV	scikit-learn, XGBoost, SHAP, pandas, NumPy, Hugging Face, ONNX Runtime, NER, transfer learning
Backend & Data	FastAPI, Pydantic, SQLAlchemy, PostgreSQL, pgvector, Qdrant, REST APIs, JWT auth, RBAC
Frontend & DevOps	React, Next.js, Tailwind CSS, Docker, GitHub Actions (CI/CD), Git, pytest, Prometheus

ACHIEVEMENTS

- Secured **3rd place** for an AI-based Task Scheduler at a competitive technical event; participated in multiple hackathons.